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Key Points:

- We developed a network representation of a synthetic earthquake catalog to train a machine-learning Machine learning (ML) model
- The trained ML model can predict the time remaining before simulated mainshocks with great accuracy
- The network representation of seismic moment and event interval enables us to accurately predict the timing of mainshocks

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

R. Norisugi,
norisugi.reiju.77e@st.kyoto-u.ac.jp

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Author Contributions:

Conceptualization: Yoshihiro Kaneko, Bertrand Rouet-Leduc

Data curation: Reiju Norisugi

Formal analysis: Reiju Norisugi,

Yoshihiro Kaneko, Bertrand Rouet-Leduc

Funding acquisition: Yoshihiro Kaneko, Bertrand Rouet-Leduc

Investigation: Reiju Norisugi

Methodology: Reiju Norisugi,

Yoshihiro Kaneko, Bertrand Rouet-Leduc

Project administration:

Yoshihiro Kaneko, Bertrand Rouet-Leduc

Resources: Reiju Norisugi,

Yoshihiro Kaneko

Software: Reiju Norisugi,

Yoshihiro Kaneko

Supervision: Yoshihiro Kaneko,

Bertrand Rouet-Leduc

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Machine Learning Predicts Earthquakes in the Continuum Model of a Rate-And-State Fault With Frictional Heterogeneities

Reiju Norisugi¹ , Yoshihiro Kaneko¹ , and Bertrand Rouet-Leduc² 

¹Graduate School of Science, Kyoto University, Kyoto, Japan, ²Disaster Prevention Research Institute, Kyoto University, Kyoto, Japan

Abstract Machine learning (ML) has been used to study the predictability of laboratory earthquakes. However, the question remains whether or not this approach can be applied in a tectonic setting where one may have to rely on sparse earthquake catalogs, and where important timescales vary by orders of magnitude. Here, we apply ML to a synthetic seismicity catalog, generated by continuum models of a rate-and-state fault with frictional heterogeneities, which contains foreshocks, mainshocks, and aftershocks that nucleate in a similar manner. We develop a network representation of the seismicity catalog to calculate input features and find that the trained ML model can predict the time-to-mainshock with great accuracy, from the scale of decades to minutes. Our results offer clues as to why ML can predict laboratory earthquakes and how the developed approach could be applied to more complex problems where multiple timescales are at play.

Plain Language Summary Accurately forecasting the timing of an impending large earthquake is one of the main but elusive goals of seismology. Recent studies show that machine learning (ML), a powerful tool that can draw inferences from hidden patterns in data, can predict the timing of laboratory earthquakes that mimic faulting in nature. However, the question remains whether or not an ML approach can be applied to natural faults where one may have to rely on sparse earthquake catalogs, and where important timescales vary by orders of magnitude. In this study, we develop and apply an ML approach to a synthetic earthquake catalog generated by physics-based computer simulations. We find that the trained model can predict the time remaining before the mainshock with great accuracy, from the scale of decades, long before the upcoming earthquake, down to the scale of hours to minutes, right before the mainshock. Our results provide clues as to why ML can predict laboratory earthquakes and how to apply an ML-based approach to real earthquakes where multiple timescales are at play.

1. Introduction

Recent studies demonstrated that machine learning (ML) could predict fast and slow slip events on laboratory faults via instantaneous monitoring of continuous acoustic emission (AE) occurring between main failure events (e.g., Borate et al., 2023; Hulbert et al., 2019; Jaspersion et al., 2021; Johnson et al., 2021; Karimpouli et al., 2023; Laurenti et al., 2022; Lubbers et al., 2018; Pu et al., 2021; Rouet-Leduc et al., 2017; Shokouhi et al., 2021). For example, Rouet-Leduc et al. (2017) showed that the statistical characteristic of AEs, notably the evolution of their variance, provides predictability of the time-to-failure in a double direct shear-experiment. Since these ML approaches rely only on real-time, continuous foreshock-like AE observations, they can provide additional information compared to classical approaches based on the recurrence interval of past large events (e.g., Savage, 1994; Schwartz & Coppersmith, 1984).

Despite its successes, notably with laboratory earthquakes, there are at least two major challenges in applying these ML approaches to forecast the timing of earthquakes in nature. First, the timescales of earthquake cycles vary by orders of magnitude, with typical recurrence intervals of large earthquakes being hundreds to thousands of years while the timescales of months to days need to be considered for short-term forecasting to be meaningful. Second, it remains unclear why ML approaches can predict time-to-failure for laboratory earthquakes, making the upscaling to real Earth problems challenging. Given that the variance and amplitude of AEs are related to the preparatory phase of the main slip event (Rouet-Leduc et al., 2017), ML might have captured hidden patterns related to the evolution of certain foreshock characteristics (e.g., Bolton et al., 2021; W. Goebel et al., 2013;

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Bertrand Rouet-Leduc
Visualization: Reiju Norisugi
Writing – original draft: Reiju Norisugi
Writing – review & editing:
Reiju Norisugi, Yoshihiro Kaneko,
Bertrand Rouet-Leduc

Johnson et al., 2013; Kwiatek et al., 2014; Lei et al., 2018; Main et al., 1989; McLaskey & Lockner, 2014; Ohnaka & Mogi, 1982; Rivière et al., 2018; Scholz, 1968; Yamashita et al., 2021).

One way to overcome these challenges is to apply an ML approach to a synthetic earthquake catalog generated by earthquake-sequence simulations in which the timescales vary by orders of magnitude and the mechanism of the earthquake generation is known or understandable (e.g., Beall et al., 2022; Cattania & Segall, 2021; Dublan- chet, 2022; Dublan- chet et al., 2013; Ito & Kaneko, 2023; Luo & Ampuero, 2018; Schaal & Lapusta, 2019; Skarbek et al., 2012; Yabe & Ide, 2017). For example, Ito and Kaneko (2023) found that in continuum models of a rate-and-state fault with frictional heterogeneities, the slope of a Gutenberg-Richter magnitude-frequency rela- tion, referred to as a b -value, decreases with time toward the mainshock, consistent with the characteristics of seismicity before large earthquakes (e.g., Gulia et al., 2016; Nanjo et al., 2012; Papadopoulos et al., 2010; Schurr et al., 2014; Shimbaru & Yoshida, 2021; Simon et al., 2021). In their model, a temporal decrease in the b -value is caused by the reduction of rupture barrier effectiveness in creeping fault patches. Applying an ML approach to this type of synthetic catalog may enable us to understand the reasons behind why and how an ML model can predict the timing of mainshocks in previous laboratory studies.

In this study, we apply ML to a synthetic seismicity catalog generated by the fault model of Ito and Kaneko (2023). The synthetic catalog contains foreshocks, mainshocks, and aftershocks that nucleate in a similar way, making it difficult to assess which output parameters contain key information for the mainshock predict- ability (Ito & Kaneko, 2023). In the following, we describe a network representation of the earthquake catalog to train an ML model (Section 2). We describe the results (Section 3) and discuss the implications (Section 4).

2. Materials and Method

2.1. Synthetic Earthquake Catalog Generated by a Rate-And-State Fault Model With Frictional Heterogeneities

The fault model that we use to generate a synthetic earthquake catalog contains a 2,400-m long, 1D fault embedded into Earth's crust, loaded by slow tectonic loading (Figures 1a–1c). The numerical approach is based on a boundary integral method (Kaneko et al., 2016; Lapusta et al., 2000), adapted for a two-dimensional (in-plane, Mode II), fully dynamic model of earthquake cycles. The constitutive response of the fault is governed by the rate-and-state friction laws with the aging form of state variable evolution (Dieterich, 1978, 1979; Ruina, 1983). The fault is characterized by regularly alternating patches of velocity-weakening (VW) and velocity-strengthening (VS) friction (Figure 1c). The details of the model setup and parameters are described in Text S1 and Table S1 in Supporting Information S1.

We use the catalog generated by the simulation during 900 years (~18,000 earthquakes). The earthquake is characterized by its epicenter x , its origin time t , and its seismic moment M (see details in Text S2 in Supporting Information S1). We categorize mainshocks as earthquakes that rupture the entire fault and all the other earth- quakes as foreshocks. For simplicity, we do not distinguish between foreshocks and aftershocks, because of the relatively few aftershocks in this model. Both the mainshocks and foreshocks nucleate in a similar manner within one of the VW patches, as a VW patch is set slightly smaller than a critical nucleation size (Rubin & Ampuero, 2005) with the setting of small characteristic slip distance $D_{RS} = 15 \mu\text{m}$ which is within a typical range of D_{RS} in rock-friction experiments (Ikari et al., 2011; Marone, 1998). Despite the relatively simple heteroge- neities, the model results in a complex spatiotemporal pattern of earthquakes (Figures 1d–1f). Foreshocks follow a power-law magnitude-frequency relation (Gutenberg & Richter, 1949) with a magnitude completeness of 2.0 (Figure S1 in Supporting Information S1). To avoid sampling smaller events limited by the size of a VW patch and potential numerical artifacts, we remove events below the magnitude completeness from the catalog (see more details on catalog preparation in Text S2 in Supporting Information S1).

2.2. Forecasting Method Using Random Forest

To predict the time to the next mainshock in the synthetic catalog mentioned above, we apply the classical random forest (RF) ML technique, implemented in scikit-learn (Breiman, 2001; Louppe et al., 2013; Pedregosa et al., 2011) (see details in Figures S3 and S4 in Supporting Information S1). We develop a data-network rep- resentation to calculate features in the earthquake catalog. A schematic explanation of our data-network repre- sentation is shown in Figures 2a and 2b. We refer to a network as a group of a specific number of consecutive

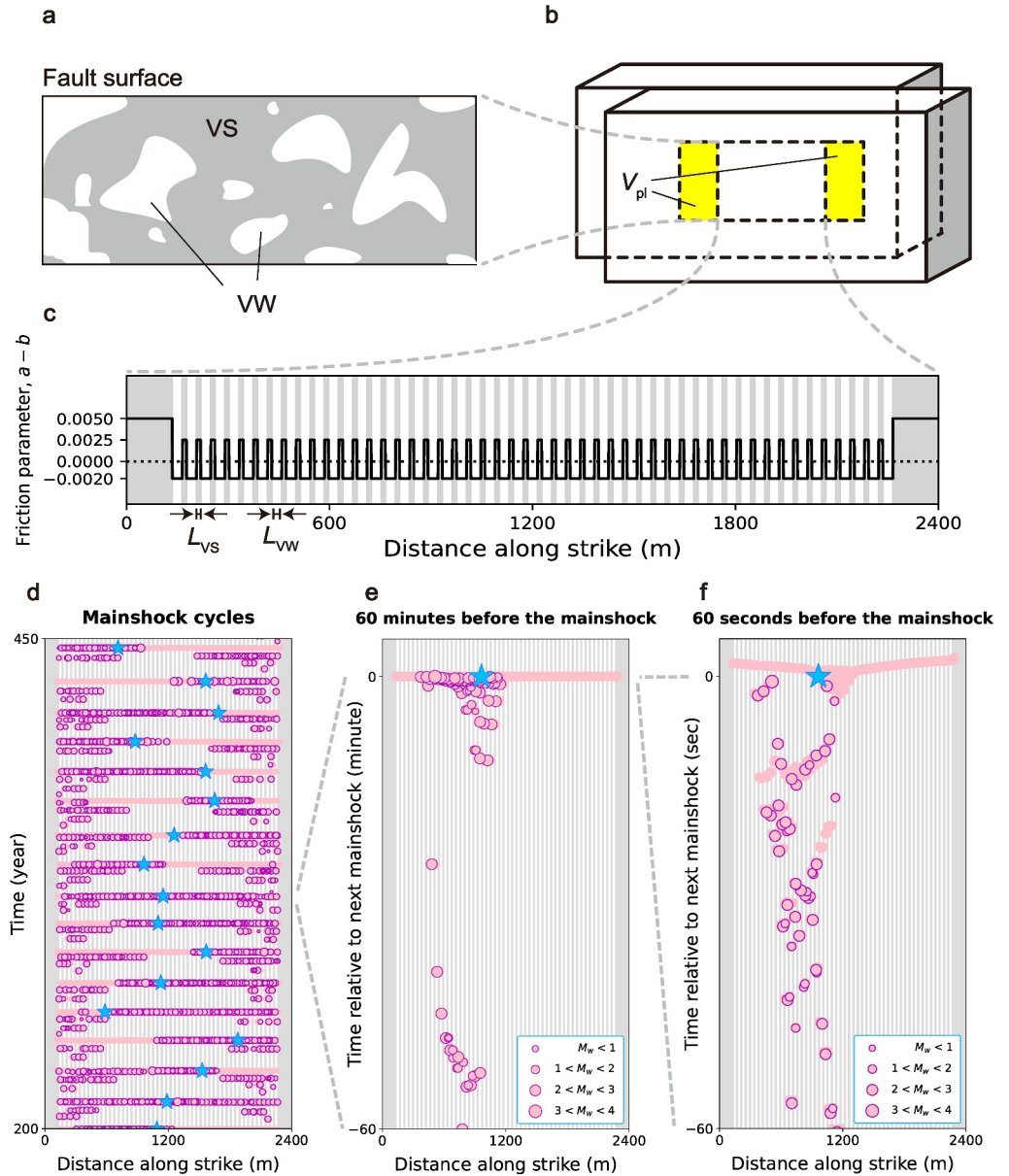


Figure 1. Fault model with frictional heterogeneities and typical active foreshock behavior in the default scenario. (a) A conceptual illustration of a fault surface with frictional heterogeneities. Gray/white areas represent velocity-strengthening and weakening (velocity-weakening) patches, respectively. (b) A schematic illustrating a strike-slip fault in the model. (c) Spatial distribution of frictional heterogeneities characterized by the rate-and-state friction parameters $a-b$ in the default scenario (d-f) Earthquake behavior over different timescales. Pink circles show the hypocenters and origin times of foreshocks and blue stars are those of mainshocks. The pink area has a higher slip velocity than 1 cm/s and indicates a coseismic slip area of individual earthquakes.

earthquakes. In the current network, we compute the changes of each earthquake pair, $\Delta x_i = x_i - x_{i-1}$, $\Delta t_i = t_i - t_{i-1}$, $\Delta M_i = M_i - M_{i-1}$, where i denotes the i th earthquake in the current network. Hence there are a total of 5 parameters that describe an earthquake pair: $X_i = (x_i, M_i, \Delta x_i, \Delta t_i, \Delta M_i)$. We compute the average and variance of X for all the pairs in a network in order to quantify the pattern of earthquake occurrence, as follows:

$$\bar{X}_j = \sum_i \frac{X_i}{j}, \quad \text{Var}(X)_j = \frac{1}{j} \sum_i (X_i - \bar{X}_j)^2 \quad (1)$$

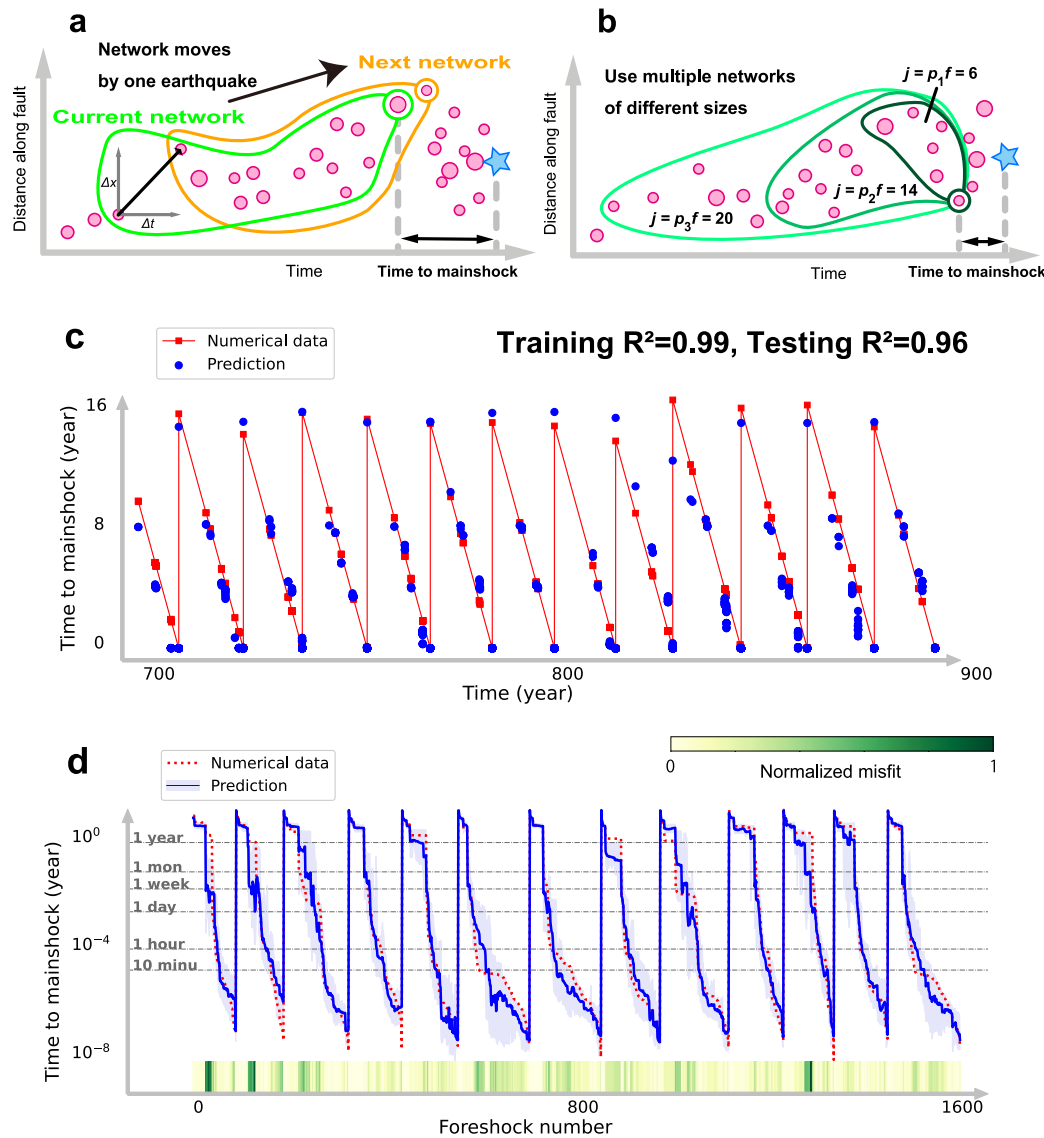


Figure 2. (a, b) Schematic of our data-network representation, showing the time and location of foreshocks (the pink circles) and one mainshock (the blue star). The green and orange foreshock groups denote the current and next networks, respectively. Statistically summarized catalog information inside the network is labeled with the time to the next mainshock, counting from the latest earthquake in the network (individually circled). (c) Prediction of the time-to-mainshock by an random forest (RF) model with multiple network representations. The red dots are the true time-to-mainshock and the blue dots are the prediction. Prediction is plotted on a year scale, corresponding to long-term prediction. Each new foreshock gives a new prediction by the RF model, irregularly spaced over time. (d) Logarithmic values of prediction as a function of the foreshock number. This corresponds to short-term prediction, down to the scale of hours to minutes. The axis of the foreshock number can be regarded as the axis of a discrete time series. The 90% confidence interval of prediction is shown in the blue color area and the normalized misfit in the logarithmic space is shown in the green color bar.

where $\bar{X}_j$ is the average, Var is the variance, and j is the number of earthquakes included in the current network, which we define as the network size. The network size j is determined as percentages of the average foreshock number $j = p f$, where p is the percentages and f is the average foreshock number in the training set. The network representation conceptually corresponds to adaptive time windows with no arbitrary duration, but with a duration that changes depending on the stage of earthquake-cycles. The percentage p is the free parameter in this method and it is optimized along with the hyperparameters of the RF model (see details in Text S3, S4, and Figure S2 in Supporting Information S1). As an example of feature notation, $\bar{M}_{5f}$ means seismic moment averaged over a

network of 5% average foreshock number. Note that all the earthquakes are considered by networks “when computing features” and there is no in-advance classification of earthquake-type (e.g., foreshocks, mainshocks, aftershocks, swarms, etc.).

Here we introduce three types of data-network representations: (a) instantaneous, (b) single, and (c) multiple network representation. The instantaneous network representation is analogous to that used in previous studies (e.g., Hulbert et al., 2019; Jaspersen et al., 2021; Johnson et al., 2021; Rouet-Leduc et al., 2017) that monitor AEs over a small time window. We define a pair of consecutive earthquakes as one network ($j = 2$), thus, the instantaneous network is the minimum one among this representation. The single network representation accounts for a specific number of earthquakes. Depending on the percentage p , the network incorporates the earthquake catalog over different timescales. As representative cases, we show three single networks, where $p = 5\%$, 50% , and 100% . The multiple network representation accounts for multiple single networks over different timescales simultaneously. A similar method is introduced by Karimpouli et al. (2023), showing the benefit of using multiple (albeit, fixed) time windows of different sizes in predicting the time-to-failure of laboratory earthquakes. The network sizes in the multiple network representation are defined as $p = (100, 90, 80, 70, 60, 50, 40, 30, 25, 20, 15, 10, 5)\%$. All the features that we eventually used are listed in Table S2 in Supporting Information S1 (see details in Text S4 in Supporting Information S1).

Labels are computed as the time remaining before the next mainshock from the latest earthquake in the current network. For networks whose latest earthquake is a mainshock, we compute the time to the next mainshock. The networks move forward each time an earthquake occurs, and the features are associated with the time of the latest earthquake in the current networks. To efficiently train the RF model on the data whose labels have many orders of magnitude (10^{-8} to 10^1 years), we change the labels to logarithmic values (see details in Text S3 in Supporting Information S1).

3. Results

3.1. The Results by the RF Model With Instantaneous, Single, and Multiple Network Representation

Figures 2c and 2d show the RF predictions based on the multiple network representation, and those via the instantaneous and single network representations are shown in Figure S3 in Supporting Information S1. We quantify the accuracy of the prediction by the coefficient of determination R^2 , where a value closer to one means the prediction is more accurate:

$$R^2 = 1 - \frac{\sum_{i=1}^n (\log_{10} y_i - \log_{10} \hat{y}_i)^2}{\sum_{i=1}^n (\log_{10} y_i - \log_{10} \bar{y})^2} \quad (2)$$

where n is the number of earthquakes, y_i is the true time-to-mainshock, $\hat{y}_i$ is the prediction, $\bar{y}$ is the average of y_i . In order to emphasize predictions at different timescales, we use logarithmic values to compute R^2 , whereby the RF approach minimizes the L2 norm cost function with the input of logarithmic labels.

The model's performances in the test set are $R_{\text{instant}}^2 = 0.72$, $R_{5\%}^2 = 0.79$, $R_{50\%}^2 = 0.87$, $R_{100\%}^2 = 0.37$, and $R_{\text{multi}}^2 = 0.96$. The instantaneous network (Figures S3a and S3b in Supporting Information S1) relies only on two earthquakes on the prediction at a given time, and hence it is highly sensitive to the recurrence interval and the seismic moment. With this simplified network as input, the RF model can roughly classify the time-to-mainshock, but cannot discriminate the detailed time-to-mainshock, leading to unstable predictions and oscillations in a broad time range. The effect of the small amount of data in the network can be seen in the single small network approach (network 5%, Figures S3c and S3d in Supporting Information S1) as well. In contrast, a much larger network size (network 100%, Figures S3g and S3h in Supporting Information S1) captures a global trend, but is insensitive to the trend toward the next mainshock and has difficulty in reliably recognizing an impending mainshock. Therefore, the medium single network (network 50%, Figures S3e and S3f in Supporting Information S1) works the best among the instantaneous and single network representations because it is in between those end-members (network 5% and 100%). In the multiple network representation, these networks of different sizes are used

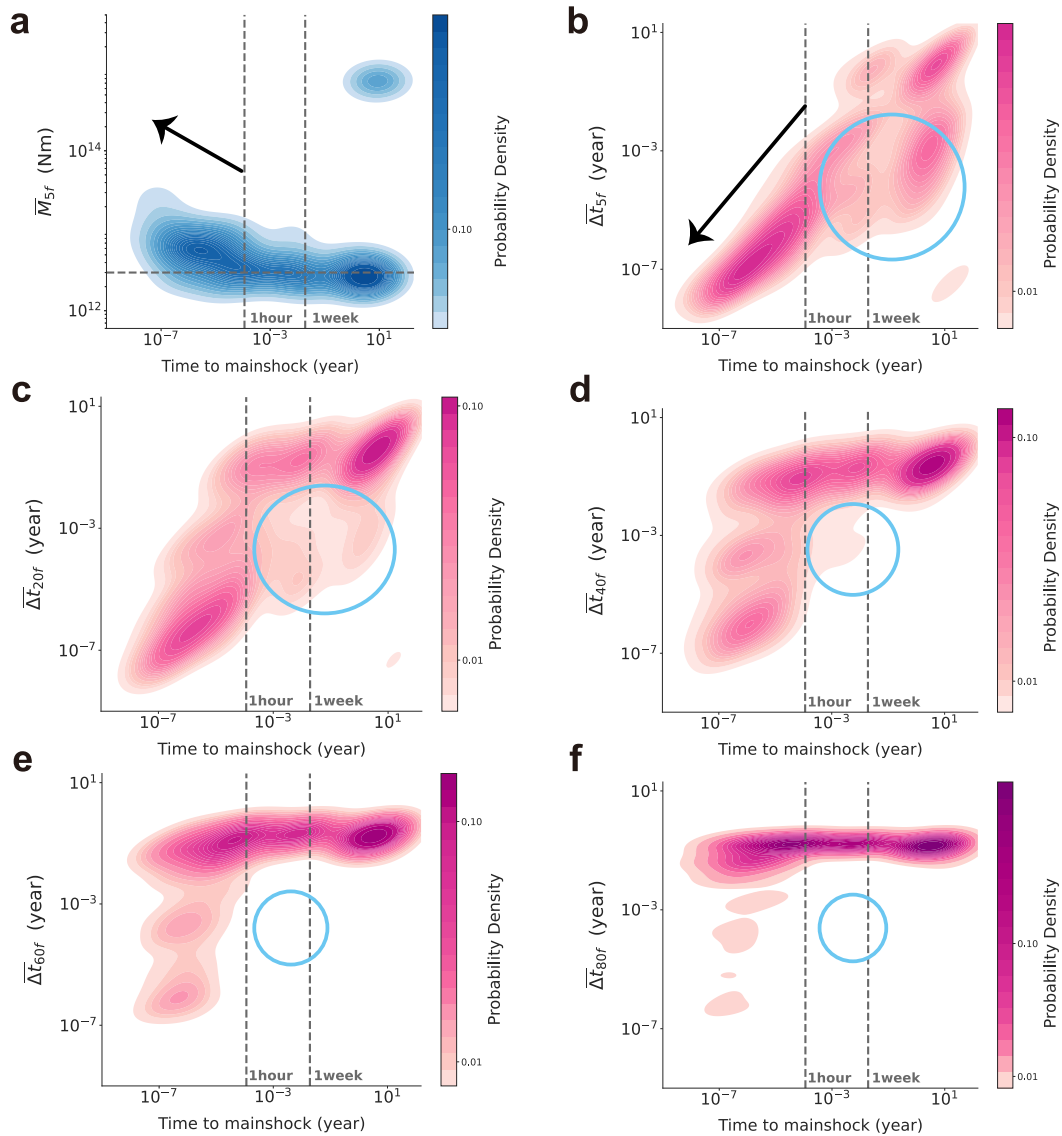


Figure 3. Feature distribution on the training set as a function of the time-to-mainshock. The contour shows the feature's probability density approximated by kernel density estimation. (a) The probability density of the average seismic moment from the network of 5% average foreshock number (b–f) The probability density of the average recurrence time from the network of 5%, 20%, 40%, 60%, 80% average foreshock number, respectively. The increase of the seismic moment and the decrease of the recurrence interval over multiple network sizes enable accurate prediction of the time-to-mainshock. Note that different features from different network sizes display predictive power at different time scales.

simultaneously, which is equivalent to using the temporal evolution of historical earthquake patterns, and the RF model can achieve a very high accuracy $R^2_{\text{multi}} = 0.96$ (Figures 2c and 2d).

3.2. Reasons Why the Multiple Network Representation Works Effectively

To explain why the multiple network representation enables accurate time-to-mainshock predictions, we visualize the temporal evolution of the features in the different network sizes (Figure 3). Figure S4 in Supporting Information S1 shows the feature importance output by the trained RF model. A small network size (e.g., network 5%) can capture the precursory behaviors, such as the increase of seismic moment and the decrease of recurrence interval when the time-to-mainshock is less than 1 hr (Figures 3a and 3b). However, they are too sensitive to mainshock-like behavior with a large number of earthquakes occurring in

a short period of time. Since the small networks contain a small number of earthquakes to compute the statistics, the data distribution becomes uncorrelated with time-to-mainshock when the mainshock is more than an hour away (a blue circled region in Figure 3b). This cluster possibly makes the RF model misunderstand that the next mainshock is approaching earlier. The large networks can add constraints in the regions where small networks have difficulty in discriminating the time-to-mainshock. Large networks contain a large number of earthquakes, which means that they tend to show precursory trends only right before the mainshocks. As shown in Figures 3c–3f, the distribution of $\overline{\Delta t}_{j=\text{large}}$ for the uncorrelated clusters (a blue circled region) tends to shrink in larger networks. When the RF model refers to these multiple networks, it accounts for the difference in feature distribution over various timescales, suppressing the oscillation of prediction and accurately predicting the time-to-mainshock. In other words, useful features present a monotonous and non-flat distribution with respect to the time before mainshock (e.g., as indicated by black arrows in Figure 3). Different network sizes have this monotonous relationship at different parts of the earthquake cycle and at different timescales, and having access to these different network sizes enables the RF model to reconcile these different timescales. In summary, the temporal evolution of the seismic moment and recurrence interval abstracted in networks of specific sizes give effective information for predicting the time-to-mainshock.

3.3. On the Accuracy of the Predictions With Training Data Size

Next, we investigate how the training data size affects the accuracy of the predictions. We generate a synthetic catalog during the 2,000 simulation years in the default parameter setting and quantify the accuracy of prediction with varying training set sizes. Specifically, we change the size of the training set from 25 years (2 mainshock cycles) to 1,800 years (115 mainshock cycles) and predict the same test set in Section 3.1 via the multiple network representation. The hyperparameters and free parameters in the RF model are determined with the multiple network approach as in Section 3.1 because these hyperparameters do not affect the result in orders. Comparing the scores of the default scenarios and the end-members, the $R^2_{\text{multi}} = 0.96$ (default scenario), the $R^2_{2 \text{ cycles}} = 0.68$, and the $R^2_{115 \text{ cycles}} = 0.97$. As expected, the smaller the training set becomes, the smaller the R^2 becomes (Figure 4a). Interestingly, the R^2 with the large training set is not largely improved even if we use about three times larger training set (115 mainshock cycles) than the default data size (41 mainshock cycles). We interpret this stagnation of the scores as the RF model reaching its upper limit and the randomness in the model preventing its accuracy from being higher. In addition, the prediction in the smallest training set (only 2 mainshock cycles) performs satisfactorily with the $R^2_{2 \text{ cycles}} = 0.68$ (Figure 4a). Despite poor performance for prediction at long time scales (Figure 4b), the trained ML model can reasonably predict the time-to-mainshock starting several weeks prior to subsequent mainshocks (Figure 4c). The effect of data size is further examined in Section 3.4.

3.4. Application to the Synthetic Earthquake Catalog Containing Aperiodic Mainshock Cycles

To understand the general applicability of the developed multiple network representation, we perform an earthquake-cycle simulation with time-varying loading. Every time the mainshock occurs, we change the loading rate V_{pl} randomly in the range of $1.0 < V_{\text{pl}} < 3.0$ mm/yr as well as the corresponding stressing rate in the range of $0.010 < \dot{\epsilon} < 0.030$ MPa/yr and keep these values constant until the next mainshock. All the other parameters of the earthquake-cycle simulation are the same as in Section 2.1. The recurrence interval varies in the range of about 10–35 years (Figure 4d). Despite the aperiodicity of the mainshock cycles, the trained ML model in the randomized loading scenario can accurately predict the multi-scaling time-to-mainshock $R^2 = 0.87$ (Figures 4e and 4f). We also examine the effect of limited data size using this catalog (Figure S5 in Supporting Information S1) and find that the ML model, trained on a minimal data set, exhibits a capability to reasonably predict the time-to-mainshocks immediately preceding their occurrence (Figure S5c in Supporting Information S1). Thus, the result indicates that the aperiodicity does not affect the performance of our ML approach, especially right before the next mainshock, supporting our conclusion that the characteristics of features mentioned in Section 3.2 can successfully capture the trend toward the next mainshock time.

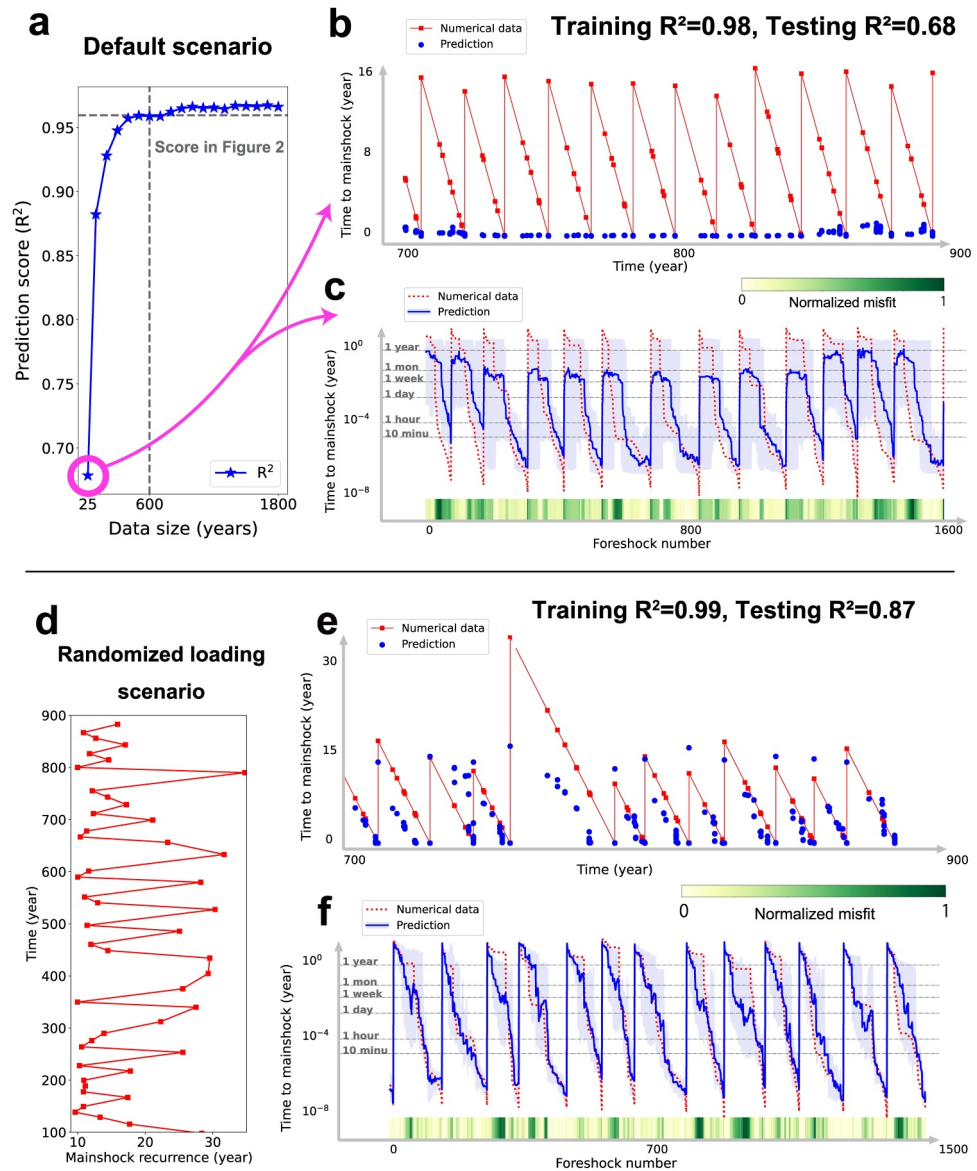


Figure 4. Performance of our approach in case of little training data and case of aperiodic earthquakes. (a) R^2 score as a function of training data size in the default scenario. The cross-dashed line denotes the score shown in Figure 2. After ~ 300 years (i.e., ~ 20 mainshocks) of training data, the prediction becomes very accurate (b, c) The long- and short-term prediction by the minimum training set (i.e., with two mainshock cycles only). (d) The recurrence interval of mainshocks of an earthquake-sequence model scenario with randomized loading rates (e, f) The long- and short-term predictions by the random forest model with multiple network representation in the randomized loading scenario.

4. Discussion

4.1. Implications for the ML-Based Prediction of Laboratory Earthquakes

From the present results, we provide a potential explanation as to why an ML approach can predict the time-to-failure in laboratory friction experiments. Our results suggest that the evolutions of important features are the increase of average seismic moment and the decrease of average recurrence interval, which is equivalent to the increase in the event's signal amplitude and frequency, consistent with the corresponding b -value decreases with time-to-mainshock (Ito & Kaneko, 2023). In the model of Ito and Kaneko (2023), the temporal decrease of the b -value emerges due to increasing shear stresses in the localized creeping patches (Figure S6 in Supporting Information S1). If we interpret laboratory AE events as tiny foreshocks, our results are consistent with the

laboratory results in which the increasing signal amplitudes or seismic moment provides the predictability of mainshocks (e.g., Rouet-Leduc et al., 2017). By inference, the reason why an ML approach can predict the time-to-failure in rock-friction laboratory experiments may be related to a decrease in b -value (Rivière et al., 2018) and corresponding changes in shear stresses in the localized creeping patches.

Our results also suggest the importance of an ML approach to account for the variations in timescales. Comparing the results by a single network and multiple networks (Figure 2 and Figure S3 in Supporting Information S1), the prediction accuracy by multiple networks is dramatically superior to the prediction by only the most important network (network 50%). Similar inferences were made in the ML results of forecasting laboratory earthquakes (Karimpouli et al., 2023). These results indicate that explicitly teaching the time evolution of features to an ML model in multiple time windows would greatly improve the prediction accuracy for laboratory earthquakes. In addition, our network representation can focus on the timescales in which precursory behavior occurs, and thus, the implementation of the multiple network representation with the ML technique would improve the prediction accuracy for laboratory earthquakes.

4.2. Toward Application to a More Realistic Earthquake Catalog

Based on the results presented so far, we discuss potential challenges when applying this technique to a real earthquake catalog. The first and most significant problem is the lack of data. Our method requires at least two mainshock cycles to train the ML model. However, there is often insufficient data or past-earthquake information to be used to train an ML model, as real earthquake catalogs are up to several decades, shorter than the recurrence intervals of typical large earthquakes. A potential solution is a transfer learning technique, which has been applied to rock-experiment data in Wang et al. (2021) by combining numerical simulation and experimental data to reduce the experimental data size needed for training. Second, our earthquake-cycle model contains a single fault and does not produce off-fault seismicity. Whether our ML approach would work for a more complex synthetic catalog needs to be tested in the future. Third, there are many different types of earthquake behaviors in nature such as mainshocks with few or no observed foreshocks, earthquake swarms, aftershocks, low-frequency earthquakes, etc., making accurately predicting the timing of mainshocks challenging.

As examples of more complex cases, we test our method with different earthquake-cycle simulation parameters, for two end-member cases: a scenario with very few foreshocks, and a swarm-like scenario (Ito & Kaneko, 2023). Specifically, in the scenario with fewer foreshocks, we decrease the friction parameter $a-b$ on the VS patches from 0.0025 (default scenario) to 0.0020, and in the swarm-like scenario, we increase them to 0.0031. The details in catalog processing are documented in Text S5 in Supporting Information S1.

In the case of the scenario with fewer foreshocks, the RF model via multiple network representation can predict the time-to-mainshock accurately with $R^2 = 0.89$ (Figure S7 in Supporting Information S1). However, to achieve this high score, we need to use a very large training data set (1,800 years earthquake-cycle, containing 105 mainshock sequences) and hence, the effect of the lack of data size is more significant, compared to the default scenario (Figure S8 in Supporting Information S1). In the case of the swarm-like scenario, the testing score is quite poor and barely above a random prediction at $R^2 = 0.10$ (Figure S9 in Supporting Information S1). The swarm-like scenario has much more complex behavior where a cluster of small earthquakes occurs right before large events that are not large enough to be categorized as the mainshocks, resulting in false-positive errors in the prediction (Figure S9 in Supporting Information S1). This is related to the fact that continuously distributed magnitudes in all the earthquakes in this scenario make the prediction task ill-defined (see details in Text S5 in Supporting Information S1). Therefore, the effort to reduce the required training data size and to develop a method dealing with more complex behaviors (including defining a mainshock in swarm-like scenarios) would be important, and testing our method in a more complex setting will be the topic of future work.

5. Conclusions

We have developed a machine-learning approach with a multiple network representation and applied it to a synthetic earthquake catalog generated by a continuum model of a rate-and-state fault with frictional heterogeneities. We have found that the ML model with multiple network representation can predict the time remaining before the mainshock with great accuracy, from the scale of decades, long before the upcoming earthquake, down to the scale of hours to minutes, right before the mainshock. We find that quantifying the occurrence patterns of earthquakes by abstracting a catalog in a specific number of groups of earthquakes is key for accurate prediction at

all timescales. The important earthquake features for predicting the timing of mainshocks are the evolution of seismic moment and recurrence interval of foreshocks averaged over multiple foreshock network sizes.

Data Availability Statement

The numerical data used to produce all the figures are available in a permanent data repository (Norisugi et al., 2024).

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Supporting Information for “Machine learning predicts earthquakes in the continuum model of a rate-and-state fault with frictional heterogeneities”

Reiju Norisugi¹, Yoshihiro Kaneko¹, Bertrand Rouet-Leduc²

¹Graduate School of Science, Kyoto University, Japan

²Disaster Prevention Research Institute, Kyoto University, Japan

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Text S1: Model setup of two-dimensional, fully-dynamic earthquake-cycle simulations

The setup of our earthquake-cycle model has been described in Ito and Kaneko (2023). We simulate earthquake cycles on a one-dimensional, planar strike-slip fault embedded into an elastic continuum, being imposed slow tectonic loading. The numerical approach is based on the boundary integral method (Lapusta et al., 2000), adopted for two-dimensional (in-plane, ModeII) fully-dynamic problems (Kaneko et al., 2016). The fully-dynamic approach enables us to simulate realistic dynamic ruptures, which would be important in physically realizing earthquake rupture arrest and hence the size of simulated earthquakes. (Lapusta et al., 2000; Ito & Kaneko, 2023).

The fault length is 2400 m and 1200 m regions at both ends of the fault are loaded by the imposed loading rate $V_{pl} = 2.0$ mm/yr. To avoid loading only from the sides of the fault, we additionally apply a time-independent stress rate $\tau_0 = 0.020$ MPa/yr uniformly along the fault. The number of cells is 16384 for the entire fault, and the cell size is 0.15 m.

The fault constitutive response is governed by laboratory derived, rate-and-state friction laws with the aging form of state variable evolution (Dieterich, 1978, 1979; Ruina, 1983):

$$\tau = \sigma \left[f_0 + a \ln \left(\frac{V}{V_0} \right) + b \ln \left(\frac{V_0 \theta}{D_{RS}} \right) \right], \quad \frac{d\theta}{dt} = 1 - \frac{V\theta}{D_{RS}} \quad (1)$$

where τ is the shear strength, σ is the effective normal stress, a and b are the rate-and-state constitutive parameters, V is the slip rate, f_0 is the frictional coefficient at $V = V_0$, θ is the state variable, and D_{RS} is the characteristic slip distance for slip variable evolution. The constitutive parameter $a - b$ mainly controls fault slip behavior. The region where $a - b$ is positive or negative corresponds to velocity-strengthening (VS) or velocity-weakening (VW) patches, respectively. In the default scenario (Figure 1c), VS ($a_{VS} - b_{VS} = 0.0025$) and VW ($a_{VW} - b_{VW} = -0.0020$) patches alternate at a fixed length of $L_{VS} = 12$ m and $L_{VW} = 30$ m, respectively. The 130-m VS regions at both ends of the fault to act as permanent rupture barriers. The number of VS and VW patches is 50 and 51, respectively. We set a uniform value of $D_{RS} = 15 \mu\text{m}$, which is within a range of typical values reported in rock-friction

laboratory experiments (Marone, 1998; Ikari et al., 2011). The setting of the parameters is summarized in Table S1.

Under slow tectonic loading, a stick-slip frictional instability develops only in the velocity-weakening region, where the instability exceeds the critical nucleation length. Theoretical estimation of the nucleation length relevant to the present simulation is

$$h_{\text{RA}}^* = \frac{2}{\pi} \frac{\mu' b_{\text{VW}} D_{\text{RS}}}{\sigma (b_{\text{VW}} - a_{\text{VW}})^2} \quad (2)$$

(Rubin & Ampuero, 2005) where σ is the effective normal stress, $\mu' = \mu/(1 - \nu)$, μ is the shear modulus, ν is the Poisson's ratio, and $a_{\text{VW}} - b_{\text{VW}}$ is the frictional constitutive parameters on the velocity weakening patch. In the parameters shown in Table S1, the estimated critical nucleation lengths are $h_{\text{RA}}^* = 31$ m. The actual critical nucleation length is affected by the background loading rate and the way the VW patch is loaded (Rubin & Ampuero, 2005; Kaneko et al., 2016). Since dynamic rupture spontaneously nucleates from VW patches in our simulation, and hence the actual nucleation size is smaller than h_{RA}^* .

Text S2: Preparation of synthetic earthquake catalog in the default scenario

In the default scenario, the dataset consists of the synthetic catalog during the 900 simulation years. We delete the first 4 mainshock cycles (about 100 years) to eliminate artificial foreshock behavior due to arbitrarily chosen initial conditions (Ito & Kaneko, 2023). Before computing the features, we split the data set into training and test sets to avoid mixing. The training set ranges from about 100 years to 700 years and the test set ranges from about 700 years to 900 years. Each set contains 41 and 13 mainshock cycles, respectively. For each mainshock cycle, the foreshock number ranges from 197 to 539. A synthetic earthquake catalog is produced as follows. We define seismic slip (i.e., an earthquake) when the slip rate exceeds 1 cm/s at any location on the fault. Following Ito and Kaneko (2023), each earthquake in the catalog is characterized by its epicenter x , its origin time t and its seismic moment $M = \mu AD$, where μ is the shear modulus, A is the square of slip length, and D is the amount of co-seismic slip. The moment magnitude of each earthquake is calculated via $M_w = \frac{2}{3}(\log_{10} M - 9.1)$. To avoid the potential artifact in generating small earthquakes in this simulation, we delete earthquakes whose magnitude is lower than the magnitude completeness in the Gutenberg-Richter magnitude-frequency relation. As shown in Figure S1, the magnitude completeness is around $M_w = 2.0$ in each earthquake-cycle scenario, and hence we use the cut-off magnitude of $M_w = 2.0$ in all the scenarios. We examine the effect of the cut-off magnitude on the ML prediction in the default scenario by changing the cut-off magnitude from $M_w = 0.0$ to 2.1. We find that the cut-off magnitude ranging from $M_w = 0.6$ to 2.1 does not significantly affect the results. When the cut-off magnitude is smaller than $M_w = 0.5$, the prediction score starts to decrease. When the cut-off magnitude is larger than $M_w = 2.2$, the number of available earthquake data becomes quite small, making it difficult to conduct the ML prediction. With the cut-off magnitude $M_w = 2.0$, the average foreshock number in the default scenario is $f = 116$.

Text S3: Random Forest overview

Machine Learning (ML) is a discipline related to statistics that seeks the relationship between input features and output labels and ensures generalization by training a model and then testing it on unseen data. In particular, the Random Forest (RF) algorithm can flexibly fit nonlinear functions, and it is widely used as a classical ML model. We use the RF algorithm (Breiman, 2001) implemented in scikit-learn (Pedregosa et al., 2011; Louppe et al., 2013). One of the purposes is to extract and interpret important features for the prediction and this is the reason why we do not focus on using more complex and advanced algorithms (i.e. deep learning). Here, we present the overview of the RF algorithm, and the details of the RF model can be seen in the next section Text S4. The RF model is an ensemble of decision trees which are the statistically generated classifiers or regressors. A single decision tree may overfit the data it is trained on (depending on its depth), but each tree is only trained on a subset of features and a subset of data points. The resulting ensemble of decision trees cannot overfit the data in the usual sense that it would hurt testing performance, and can be generalized to the testing data (Breiman, 2001).

A single tree is generated statistically by the following operations. At the beginning, we are at the root node, and the node asks the question "Is feature k above/below the threshold t_k ?". If the answer is yes/no, then we go to the next left/right node. This process splits the data set in the root node into two data sets. The next node also asks another question to further split the data sets into subsets. In each split, the threshold is defined by the Classification and Regression Tree (CART) algorithm which attempts to split the current node into 2 purest next nodes by minimizing the cost function J ,

$$J(k, t_k)_{s,j} = \frac{m_{j,\text{left}}}{m_j} \text{MSE}_{j,\text{left}} + \frac{m_{j,\text{right}}}{m_j} \text{MSE}_{j,\text{right}} \quad (3)$$

$$\text{MSE}_{j,\text{node}} = \sum_{i \in \text{node}} (\hat{y}_{j,\text{node}} - y_{j,i})^2, \quad \hat{y}_{j,\text{node}} = \frac{1}{m_{j,\text{node}}} \sum_{i \in \text{node}} y_{j,i} \quad (4)$$

where $J(k, t_k)_{s,j}$ is the cost function when the current node j is separated by split s using feature k at the threshold t_k , $m_{j,\text{node}}$ is the number of data points in the next node, m_j is the number of data points before the split, $\hat{y}_{j,\text{node}}$ is the averaged value in the next node, and $y_{j,i}$ is the label values in the next node. Minimizing the cost function $J(k, t_k)_{s,j}$ corresponds to finding the feature k and threshold t_k such that the data within the split node becomes as homogeneous as possible and the split two nodes become as inhomogeneous as possible. That is, this process corresponds to finding the linear relationship or clusters in the high-dimensional feature space. This operation is continued until we reach the leaf nodes, which are at the top of the tree, and each leaf node has possible prediction values classified by yes/no questions based on the input features. Thus, a map is built from the input features to the output labels in the training phase. The preparation of appropriate features makes the predictions better because the input features and the output labels are directly connected by the map. In the testing phase, each data point in the test set that is completely unknown to the RF model falls into a leaf node by following the map based on the feature values of the test set, and each tree can output corresponding prediction values in the leaf nodes. Finally, the RF model averages prediction values of a set of decision trees and outputs them as a prediction of the test set.

The importance of the features is defined by the weighted average of the reduction in impurity at each node. According to the above process, the sum of the variance of the split two nodes is lower than the variance of the node before the split. Then, the amount of the drop of the impurity (variance of the data in the current node) leads to the importance of the features, because the useful feature can split the dataset into two different clusters: producing a large drop of the impurity. The feature importance is defined as the mean decrease impurity (MDI) for the split (s, j) of all the decision trees T of the RF model using the feature k

127 (Louppe et al., 2013):

$$\text{Importance}(k) = \text{MDI}(k) = \frac{1}{m_T} \sum_T \sum_{(s,j) \in T: v(s,j)=k} \frac{m_j}{m} \Delta J(k, t_k) \quad (5)$$

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$$\Delta J(k, t_k) = J(k, t_k)_{s,j} - J_j \quad (6)$$

129 where m_T is the number of the data points in trees T , $v(s, j)$ is the variable that does the split
130 (s, j) , m is the number of all the data points, and J_j is the impurity before the split (s, j) .

Text S4: Random Forest details

For the work of the RF model, we must find the regularization hyperparameters to add the appropriate constraints in making the shape of the trees: `n_estimators` corresponds to the number of trees that output each prediction, `max_depth` corresponds to how many times split the data set into the subsets, `min_samples_split` corresponds to when the tree stops further splitting of the small subsets, and `max_features` corresponds to how much diverse the tree becomes. The decision trees can fit any kind of data if we do an infinite splitting of the nodes, but it leads to decreased performance of the RF ensemble of trees. To avoid decreasing performance, we regularize these 4 hyperparameters and the network sizes (the free parameter in the data-network representation) by minimizing the averaged R^2 score in the 10-fold cross-validation with grid search. Note that the R^2 score is computed by the logarithmic value of the prediction and labels, as follows, to account for the contribution of multi-scale labels at the same rate:

$$R^2 = 1 - \frac{\sum_{i=1}^n (\log_{10} y_i - \log_{10} \hat{y}_i)^2}{\sum_{i=1}^n (\log_{10} y_i - \log_{10} \bar{y})^2} \quad (7)$$

where n is the number of earthquakes, y_i is the true time-to-mainshock, $\hat{y}_i$ is the predictions, $\bar{y}$ is the average of y_i . Figure S2 shows the way to split the training set into 10 different pairs of the validation set and the training set for the validation. Note that, since the data-network representation summarizes past information in the current data point, we must keep the time series to adequately evaluate the model performance, not mix the training and validation set. The hyperparameters obtained by 10-fold cross-validation in the prediction of default scenario with multiple network representations are as follows, `n_estimators` = 250, `max_depth` = 20, `min_samples_split` = 30, and `max_features` = 10. In the instantaneous and single network approach, we use the same hyperparameters as the multiple network approach except for `max_features` = 4.

Network sizes j in the multiple network representation are also defined by the cross-validation as follows, $j = p f$, where f is the average foreshock number, and p is the percentages $p = (5, 10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100)$. Note that the networks of similar size are correlated, but the results in cross-validation get better when we use these multiple networks, and this is the reason why we add the percentages of middle numbers ($p = 5, 15$, and 25). We check that adding more networks (both smaller and larger sizes) does not significantly change the prediction scores in this study, but the choice of the number of networks would likely depend on the problem of interest.

We also optimize the features used for ML prediction based on the feature importance output by the trained ML model. In the first trial with all the features we mentioned in the main text, the spatial information x and Δx has too little importance compared to the information related to seismic moment M and recurrence interval Δt . Besides, the statistics of ΔM also have little importance. Thus, we eventually delete those features related to x , Δx , and ΔM , then use just the features from Δt and M in this study (Figure S4). All features that we used are summarized in Table S2. We confirm that deleting less-important features improves the score in the cross-validation.

Text S5: Preparation of synthetic earthquake catalog in the fewer-foreshock scenario and the swarm-like scenario

In the fewer foreshock scenario, we use 2000-year earthquake-cycle data. Decreasing the $a - b$ on VS patches suppresses stress interaction and results in a less-active foreshock scenario (Ito & Kaneko, 2023). As in the default scenario, the first 100 years are removed from the catalog. Training and test sets contain 105 and 12 mainshock cycles, respectively. The foreshock number ranges from 84 to 430. After events smaller than magnitude 2.0 are removed based on the magnitude-frequency distribution (Figure S1b), the average foreshock number in the training set is $f = 75$.

In the swarm-like scenario, we use 900-year earthquake-cycle data. Increasing the $a - b$ on VS patches promotes stress interaction and results in a swarm-like active scenario (Ito & Kaneko, 2023). Here, we refer to swarm as an earthquake scenario without clear mainshocks (e.g. inconsistent recurrence or magnitude of large earthquakes). Thus, even in the swarm-like scenario, our approach does not distinguish earthquakes (i.e. foreshocks and mainshocks) when computing features. As in the default scenario, the first 100 years are removed from the catalog. Training and test sets contain 42 and 14 mainshock sequences, respectively. The foreshock number ranges from 262 to 2097. After removing events smaller than magnitude 2.0 based on the magnitude-frequency distribution (Figure S1c), the average foreshock number in the training set is $f = 265$. This scenario produces a continuous distribution of magnitude-frequency relation (Figure S1c) and it is unclear how to define which earthquakes should be categorized as the mainshocks. Here we define relatively large earthquakes as the mainshocks such that the resulting recurrence intervals are similar to the aperiodic behavior shown in Figure 4d (Figure S9b). This scenario takes 3 months to compute a 900-year result, so further studying the swarm-like scenario requires more efficient computing of earthquake-cycle simulation.

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Table S1. Model parameters assumed in the present study

Parameters		
ρ	Density	2610 kg/m ³
μ	Rigidity	23.3GPa
ν	Poisson's ratio	0.216
V_{pl}	Loading rate	2.0 mm/yr
$\dot{\tau}$	Shear stress rate uniformly added to the entire fault	0.020 MPa/yr
σ	Effective normal stress	50 MPa
f_0	Reference frictional coefficient	0.60
D_{RS}	Characteristic slip distance for slip evolution	15 μ m
c_p	P-wave speed	5.0 km/s
c_s	S-wave speed	3.0 km/s
a_{VS}	Frictional constitutive parameter on VS patches	0.020
b_{VS}	Frictional constitutive parameter on VS patches	0.0169 – 0.018
a_{VW}	Frictional constitutive parameter on VW patches	0.020
b_{VW}	Frictional constitutive parameter on VW patches	0.022
L_{VS}	Length of VS patches	12 m
L_{VW}	Length of VW patches	30 m

Table S2. Summary of the features where network sizes are
 $j = (100, 90, 80, 70, 60, 50, 40, 30, 25, 20, 15, 10, 5)f$

Features in first trial		Optimized features	
$\bar{x}_j$	$\text{Var}(x)_j$	$\bar{\Delta t}_j$ $\bar{M}_j$	$\text{Var}(\Delta t)_j$ $\text{Var}(M)_j$
Δx_j	$\text{Var}(\Delta x)_j$		
Δt_j	$\text{Var}(\Delta t)_j$		
$\bar{M}_j$	$\text{Var}(M)_j$		
$\Delta \bar{M}_j$	$\text{Var}(\Delta M)_j$		

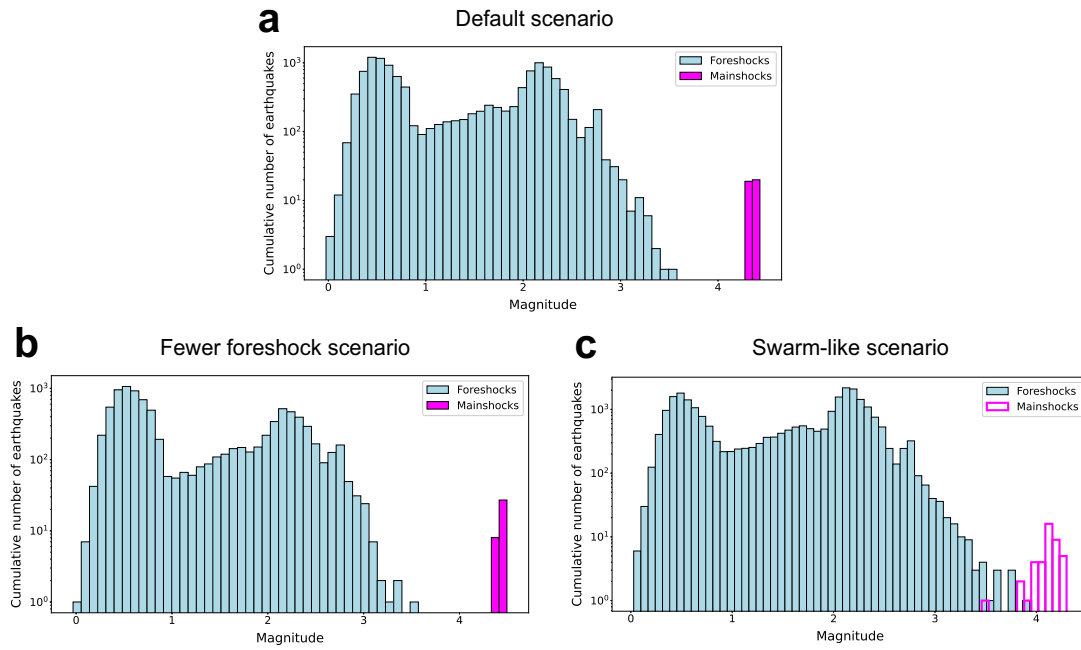


Figure S1. (a-c) The magnitude-frequency distribution in the default scenario (a), the fewer-foreshock scenario (b), and the swarm-like scenario (c), respectively. The blue and pink histograms indicate the distribution of foreshocks and mainshocks defined. The Gutenberg-Richter relation (Gutenberg & Richter, 1949) is partially reproduced with a magnitude completeness of around 2.0 in each scenario. The incompleteness of the number of small earthquakes seems to be caused by the resolution of the numerical model and the patch size of the frictional heterogeneity.

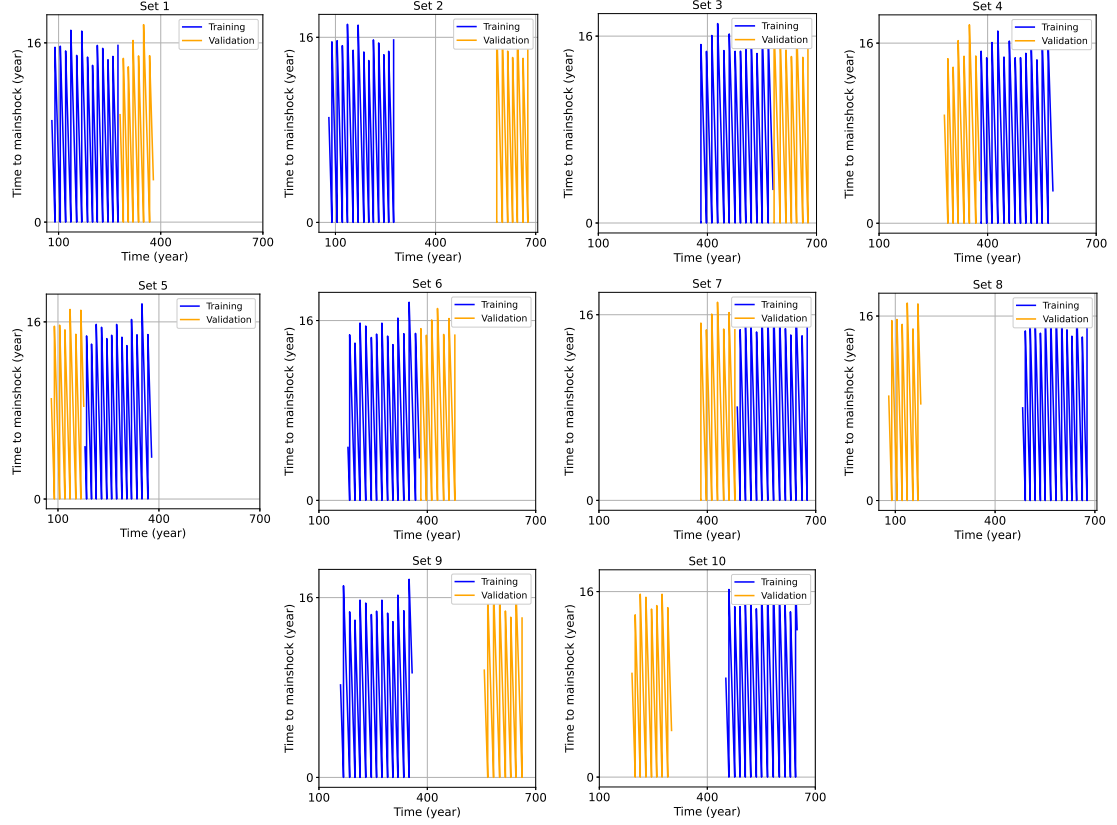


Figure S2. The 10 different ways to split the training set into the validation set (yellow) and the training set for the validation (blue). To properly estimate the model performance with data-network representation, we keep the time series of each subset. The training and validation sets contain approximately 13 and 7 mainshock cycles, respectively.

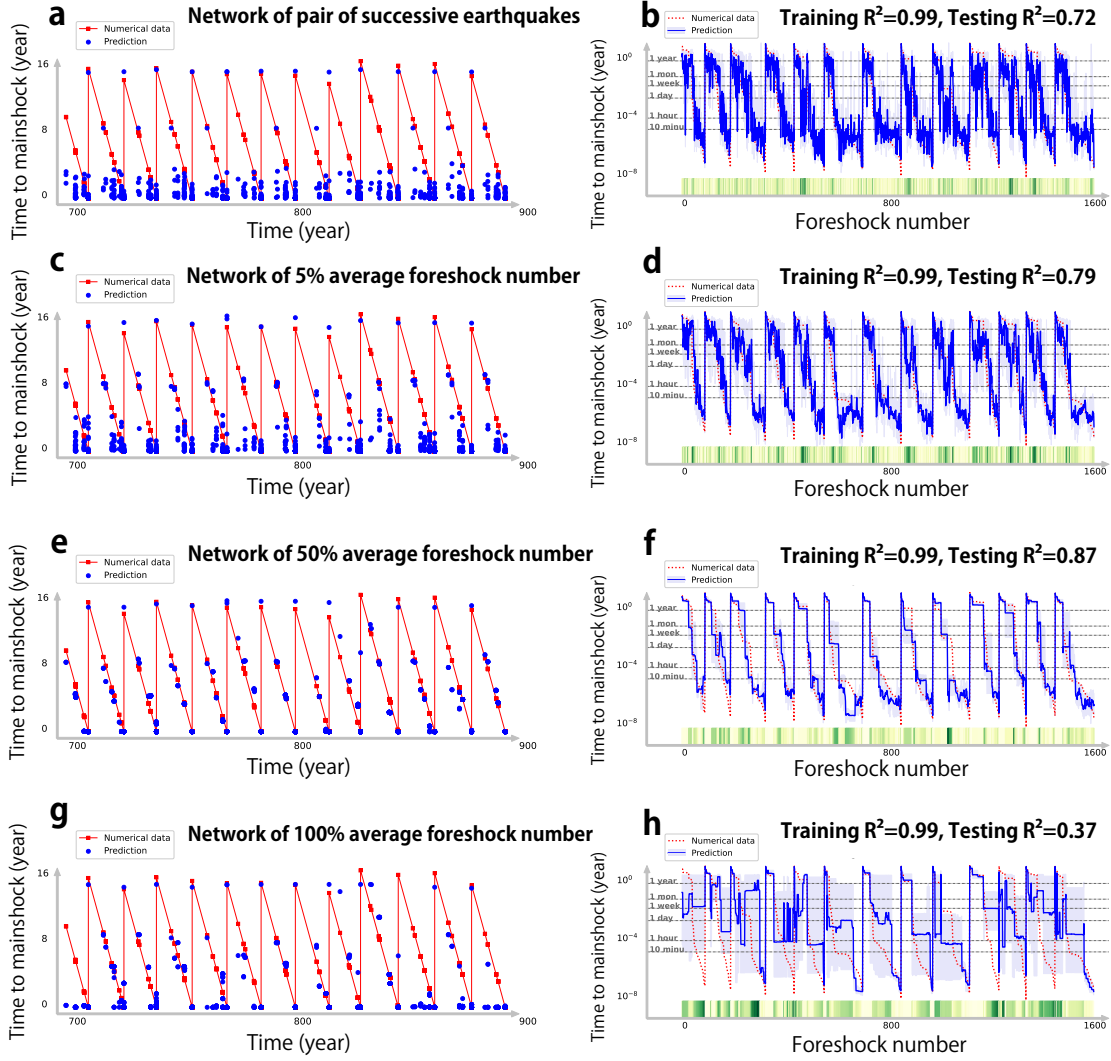


Figure S3. The property of the prediction as a function of the network size. The prediction by the instantaneous network representation (a-b), 5% single network representation (c-d), 50% single network representation (e-f), 100% single network representation (g-h). The left and right sides of this Figure correspond to the long- and short-term predictions, respectively.

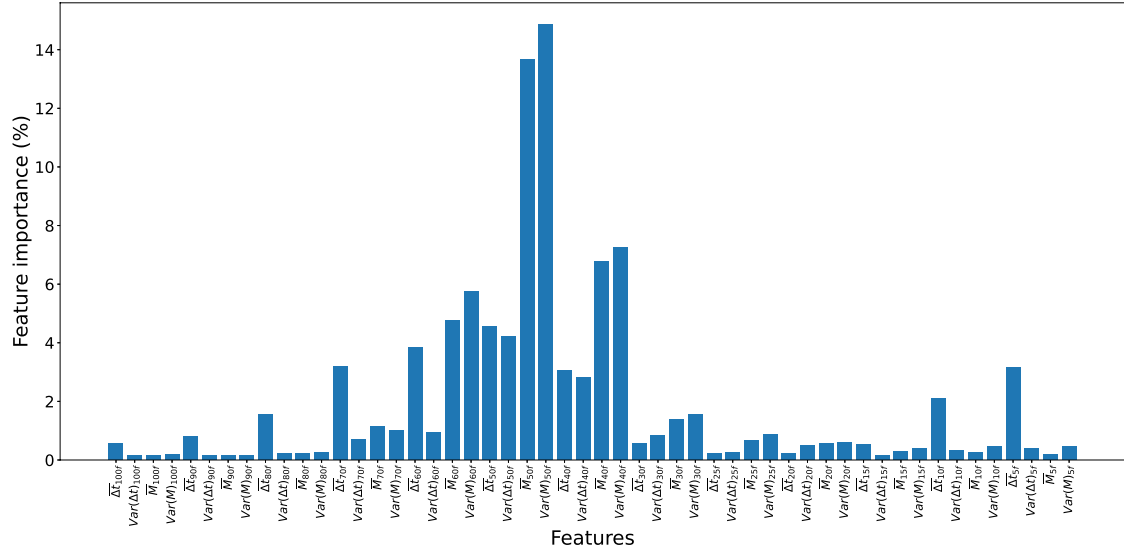


Figure S4. Feature importance of the RF model with multiple network representation. Regarding the notation of the features, for example, $\bar{M}_{5f}$ means that the seismic moment averaged over the network of 5% average foreshock number, and $\text{Var}(\Delta t)_{10f}$ means that the variance of the recurrence interval taken over the network of 10% average foreshock number. The medium size network (40% or 50% average foreshock number) is the most important.

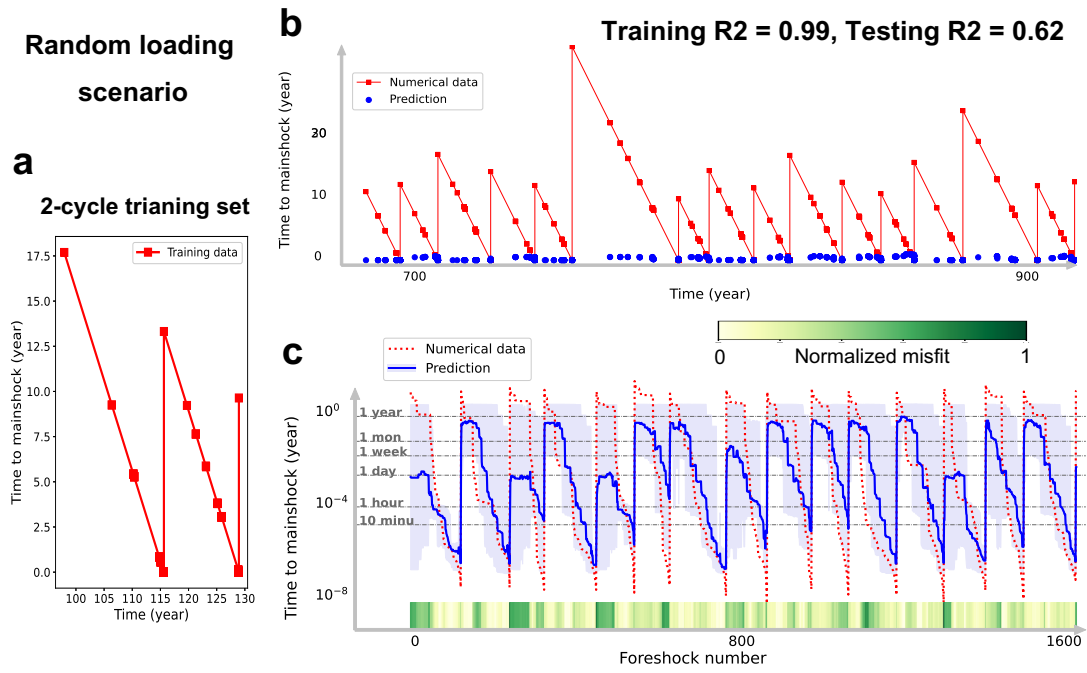


Figure S5. Prediction with 2-cycle training set in the randomized loading scenario. (a) Recurrence interval of the training set. (b-c) The long- and short-term prediction with 2-cycle training set in the randomized loading scenario. The prediction gets better when closing the next mainshocks except the mainshock recurrence is much greater than the training set.

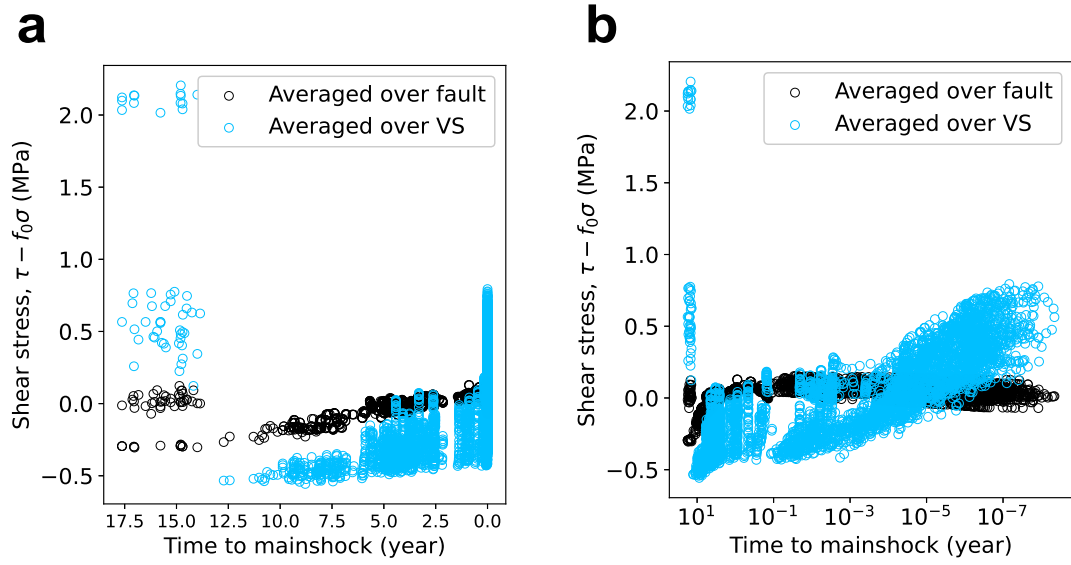


Figure S6. Shear stress evolution in the default earthquake-cycle scenario. (a) and (b) show year-scale and logarithmic timescale shear stress evolution, where $\tau - f_0\sigma$ is shear stress minus reference shear stress. The black dots indicate the shear stress averaged over the entire fault and the blue dots indicate the shear stress averaged over the VS patches (i.e., creeping patches). Each dot shows the shear stress value sampled when earthquakes occur. All the mainshock cycles in the training set (~ 100 to 900 years) are included in this figure.

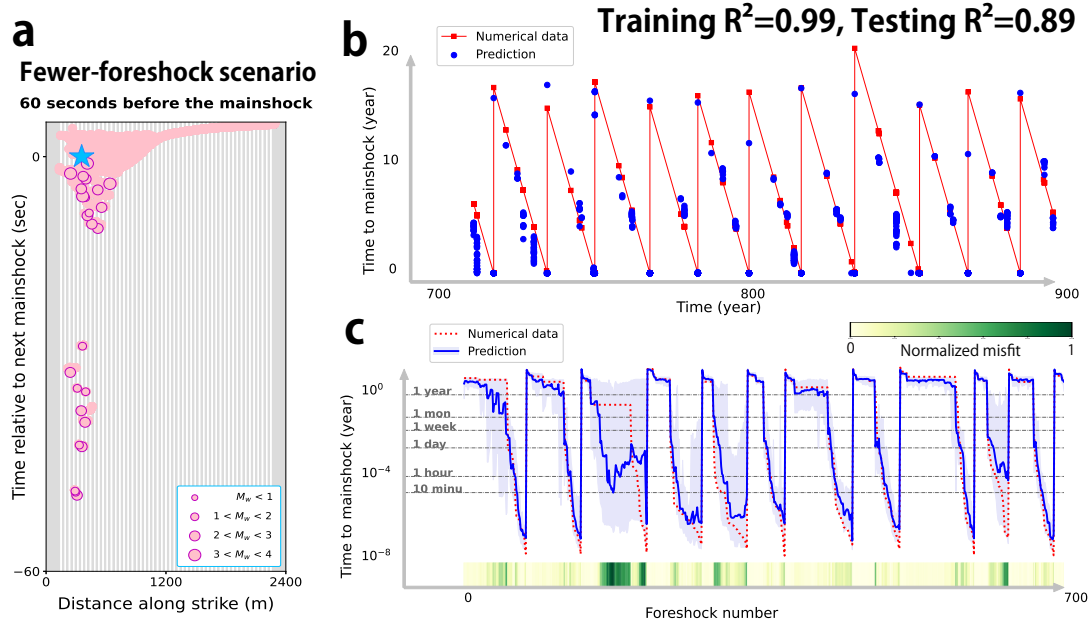


Figure S7. The prediction in the fewer-foreshock scenario ($a_{VS} - b_{VS} = 0.0020$). (a) A representative foreshock sequence just before the next mainshock. The average foreshock number ranges from 80 to 430. (b-c) The long- and short-term prediction in the fewer-foreshock scenario. To achieve this high score, we need to use a very large data set (1800 years, containing 105 mainshock sequences), compared to the default scenario (Figures 2c-d). This is due to the fewer number of foreshocks than the default scenario, making it difficult to capture the precursory trend in the individual mainshock sequence.

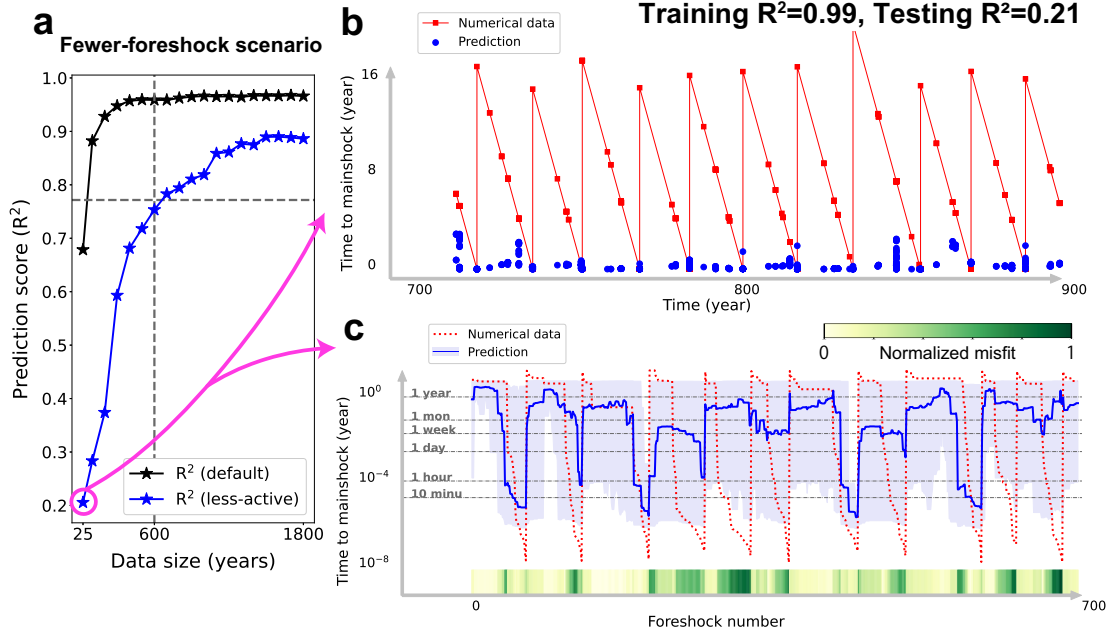


Figure S8. The effect of the data size for training in the fewer-foreshock scenario. (a) The R^2 scores as a function of the data size of the training set. We change the data size from about 25 years (contains only 2 mainshock sequences) to 1800 years (contains 105 mainshock sequences), and predict the same test set as shown in Figure S6. The black line shows the data size effect in the default scenario (Figure 4a). (b-c) The prediction of the RF model with multiple network representation by the smallest training set. Although the prediction with a small data size is accurate just before the mainshock in the default scenario (Figures 4b-c), the same approach with a small data size does not accurately predict the mainshocks in the fewer-foreshock scenario. The R^2 score reaches 0.21 with minimum training data size and the prediction has a broad range of confidence intervals, which means the prediction is not reliable.

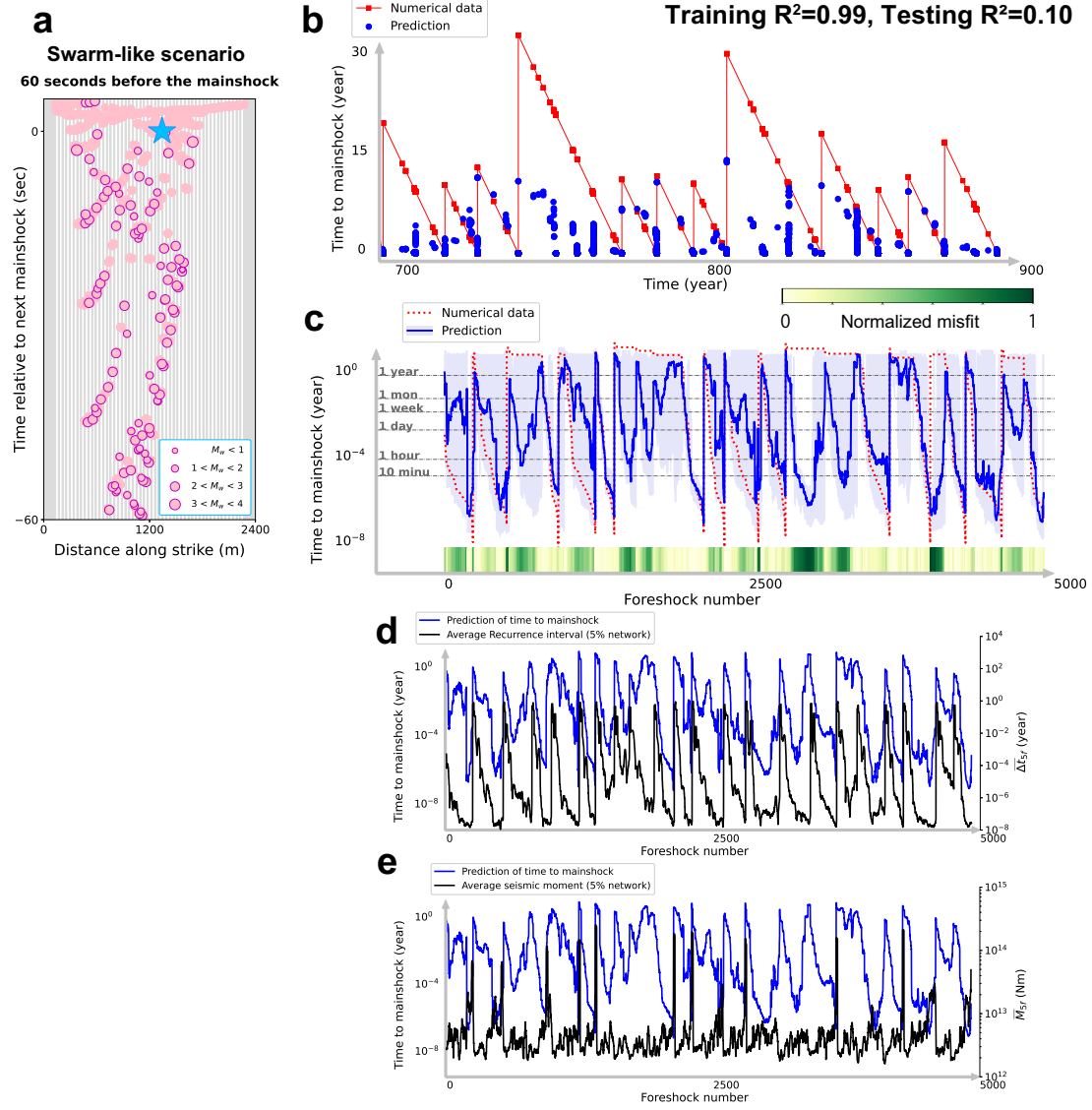


Figure S9. The prediction in the swarm-like scenario ($a_{VS} - b_{VS} = 0.0031$). (a) A representative foreshock sequence just before the next mainshock. The foreshock number ranges from 260 to 2100. (b-c) The long- and short-term prediction in the swarm-like scenario. (d-e) Comparisons between prediction versus features ($\bar{\Delta t}_{5f}$, and $\bar{M}_{5f}$). A good correspondence between the prediction for time-to-mainshocks and the average recurrence interval or seismic moment indicates that the errors in the prediction are false positives due to the ill-defined prediction task.